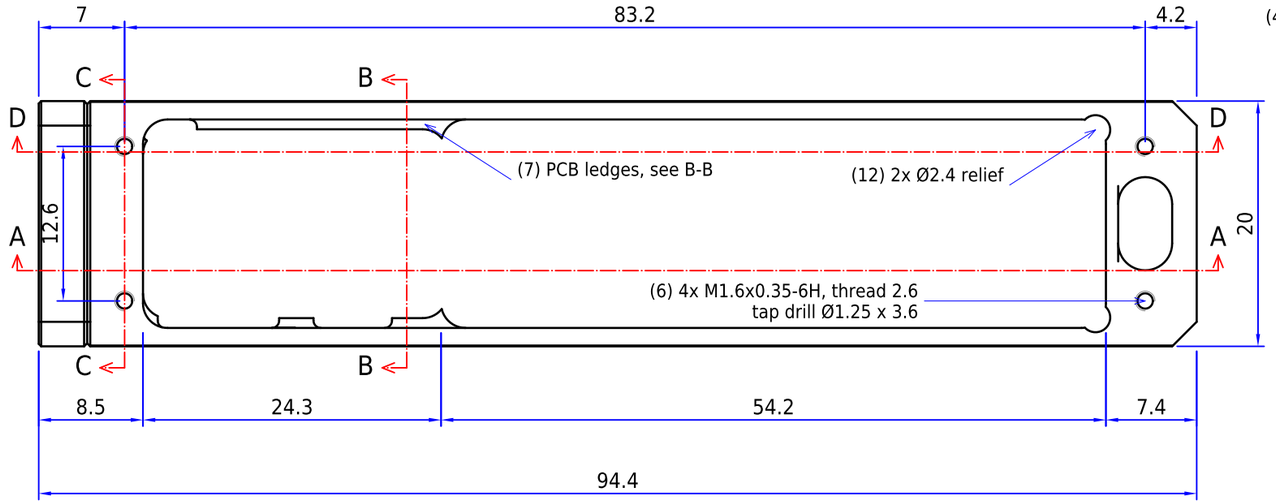
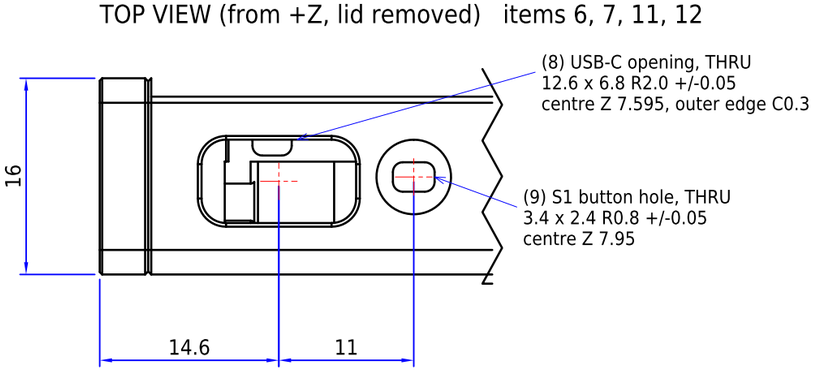
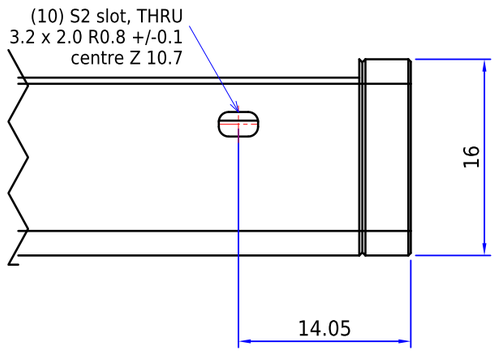


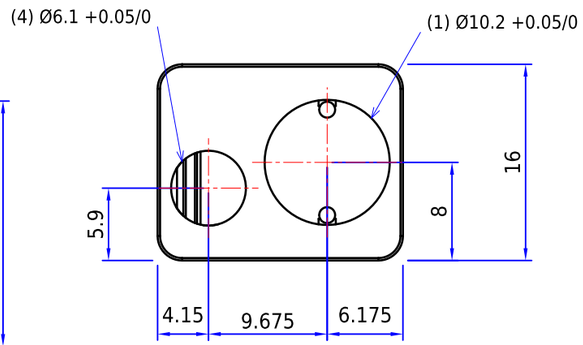
TOP VIEW (from +Z, lid removed) items 6, 7, 11, 12



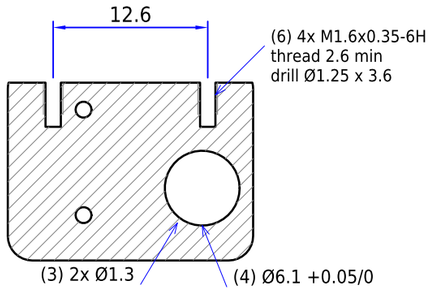
SIDE VIEW from -Y (control side) items 8, 9



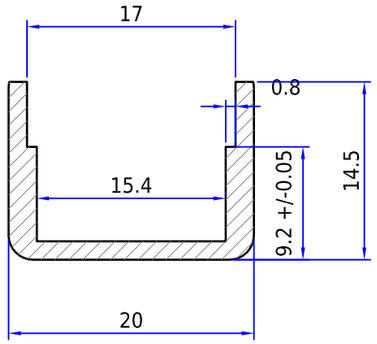
SIDE VIEW from +Y (laser side) item 10



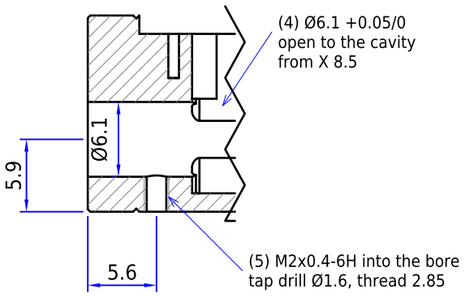
FRONT VIEW (from -X) items 1-4



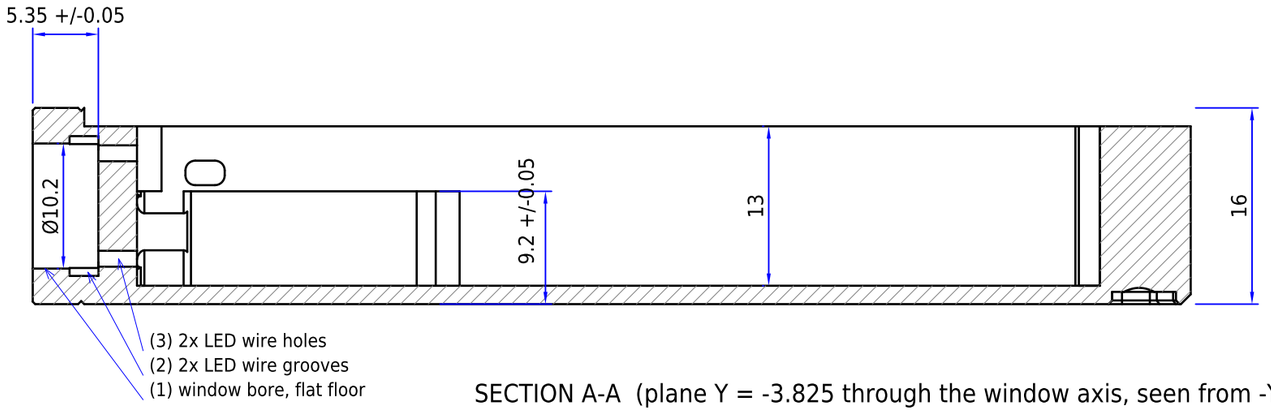
SECTION C-C (X = 7) items 3, 4, 6



SECTION B-B (X = 30) items 7, 11



SECTION D-D (Y = 5.85) items 4, 5



SECTION A-A (plane Y = -3.825 through the window axis, seen from -Y) items 1, 2, 3, 7, 11

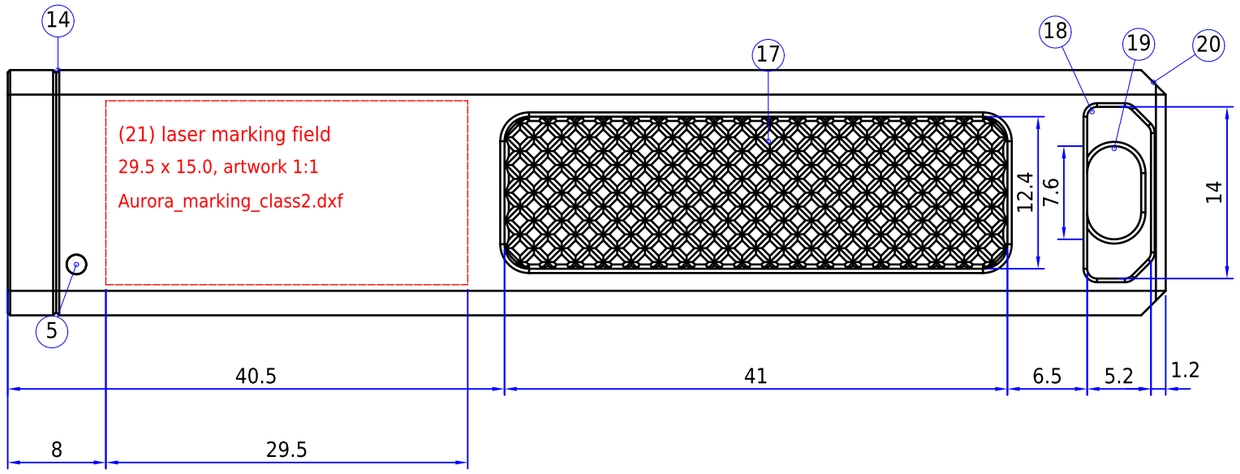
ID	FEATURE	SPECIFICATION	TOLERANCE / NOTE
1	Window bore (front)	Ø10.2 x 5.35 deep, flat floor; axis Y -3.825, Z 8.0 = 6.175 from the control-side face	Ø +0.05/0; depth +/-0.05 floor flat 0.03, Ra 1.6
2	LED wire grooves 2x	at 12 and 6 o'clock: 1.4 wide, bottom R5.7 from the axis, X 3.0 to 5.35 (bore floor)	+/-0.1
3	LED wire holes 2x	Ø1.3 from the bore floor into the cavity (X 5.35 to 8.5), at Z 8.0 +/- 4.3, Y -3.825	+/-0.1
4	Laser bore (front)	Ø6.1, axis Y 5.85 (4.15 from the +Y face), Z 5.9; open to the cavity at X 8.5 (module 6 x 12, recess 0.3)	Ø +0.05/0; axis +/-0.05
5	Set-screw hole (bottom)	M2x0.4-6H from the bottom face into the laser bore, X 5.6, Y 5.85; tap drill Ø1.6, thread length 2.85	6H; position +/-0.1
6	Lid screw threads 4x	M1.6x0.35-6H, drill Ø1.25 x 3.6 deep (flat), thread 2.6 min; X 7.0 and 90.2, Y +/- 6.3	6H; position +/-0.1
7	PCB ledges	Z 9.2, 0.8 wide, X 8.5 to 32.8. None: -Y at X 9.7 to 19.6 (USB-C) and 22.4 to 28.8 (S1); +Y at X < 12.9 (laser)	Z +/-0.05; both ledges coplanar within 0.05
8	USB-C opening (-Y wall)	12.6 x 6.8 R2.0 through; centre X 14.6, Z 7.595; outer edge C0.3	+/-0.05
9	S1 button hole (-Y wall)	3.4 x 2.4 R0.8 through; centre X 25.6, Z 7.95	+/-0.05
10	S2 slot (+Y wall)	3.2 x 2.0 R0.8 through; centre X 14.05, Z 10.7	+/-0.1
11	Cavity	17.0 wide (1.5 walls), floor Z 1.5, lid seat Z 14.5; vertical corners R2.0; trough 15.4 wide between the ledges	+/-0.1
12	Bay corner reliefs 2x	Ø2.4 through the bay-end corners, centres X 86.151, Y +/- 7.651	+/-0.1
13	Exterior, lid, marking	V-ring, side grooves, thumb dish, tail and lanyard slot, knurl, marking field: sheet 2. Lid: drawing AUR-H-02.	

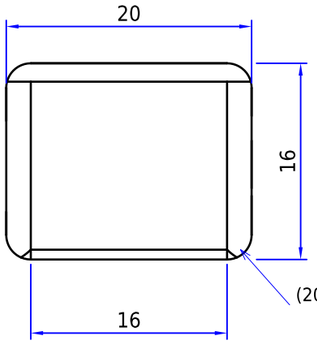
NOTES
1 Material: aluminium 6061-T6. Geometry per Aurora_tray.step; this sheet adds threads, fits, finish and notes.
2 General tolerances ISO 2768-mK; toleranced values are critical.
3 Threads are tapped before anodising and gauged after finishing.
4 Break all sharp edges 0.1-0.2. Deburr the USB-C, S1 and S2 openings from inside and the window bore edge: glass (10.0 x 1.0) and DTP board (10.0 +/-0.1) must drop in.
5 Finish: fine glass-bead blast, then anodise type II matte black (8-12 um). Tolerances apply after finishing.
6 Window bore floor (X 5.35) is the LED board contact face: flat 0.03, Ra 1.6, no burr or anodise step.
7 Thin walls in the STEP (mm): 1.075 window bore to control face; 1.1 laser bore to +Y face; 0.875 front tap to head back face; 1.025 front tap to window bore and bore floor; 1.0 tail front wall; 0.9 thumb dish to cavity; 0.825 V-ring to window bore; 1.118 tail window to bay corner (sheet 2). Finish them in a light last pass.
8 Inner vertical corners R2.0 (pocket 13.0 deep); the bay corner reliefs (12) are intended. Floor edges stay sharp (flat end mill).
9 Sections: A-A at Y -3.825 (window axis), D-D at Y 5.85 (laser axis), B-B at X 30, C-C at X 7.0 (front taps).

MAIN BODY (TRAY)

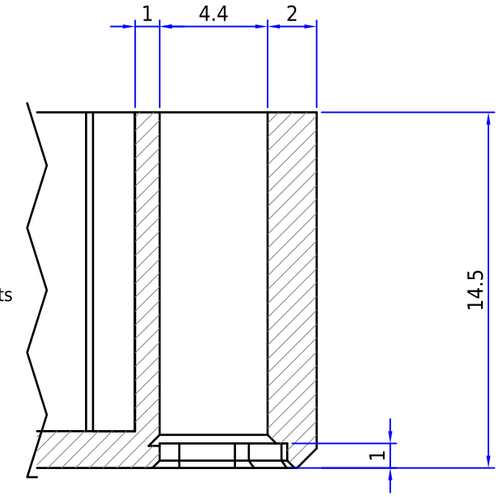
Part: AUR-H-01 Aurora_tray.step	SCALE 2:1 (details as marked) SHEET 1/2
Aurora++ keychain flashlight v0.5 2026-10-14	Units mm | ISO 2768-mK

BOTTOM VIEW (from -Z) items 14, 17-21 (5: sheet 1)

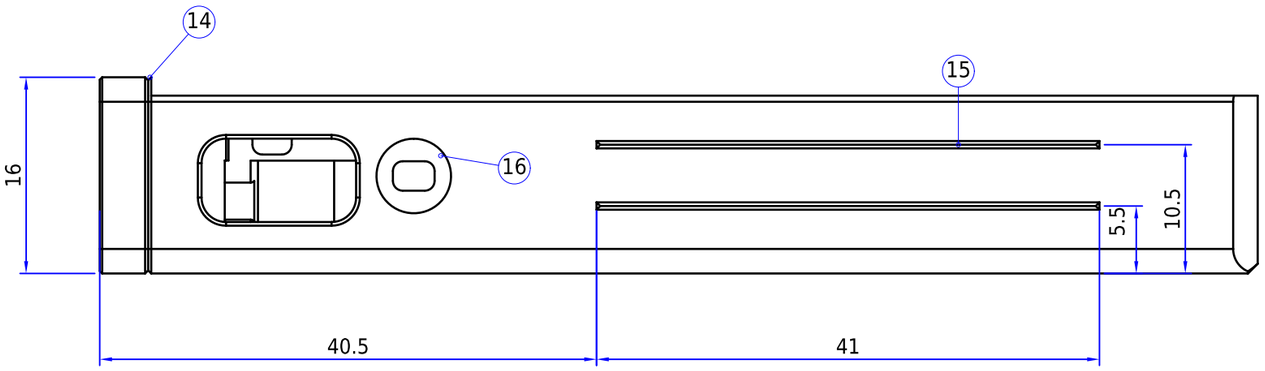




TAIL VIEW (from +X) item 20

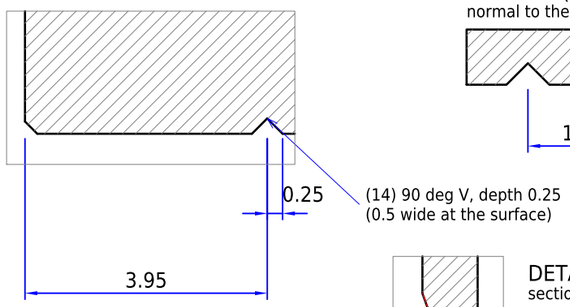


SECTION F-F (Y = 0) items 18, 19

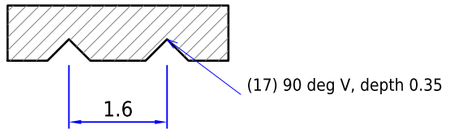


SIDE VIEW from -Y items 14, 15, 16

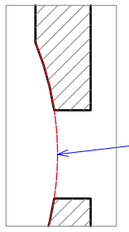
DETAIL V (10:1) V-ring, bottom face
section Y = -3



DETAIL K (10:1) knurl profile
normal to the grooves, schematic



DETAIL E-E (6:1)
section X = 25.6



ID	FEATURE	SPECIFICATION	TOLERANCE / NOTE
14	V-ring (seam groove)	90 deg V-groove ring round the body (bottom, both sides, top): apex at X 3.95, 0.25 deep, 0.5 wide at the surface; its rear flank ends at the lid front edge X 4.2 (the lid seam)	depth +/-0.05 sharp apex
15	Side grooves 4x	90 deg V-grooves in both side faces at Z 5.5 and 10.5: X 40.5 to 81.5 at the surface (apex line X 40.8 to 81.2), depth 0.3, 0.6 wide, pyramid run-outs	depth +/-0.05
16	Thumb dish (-Y face)	spherical SR8.0, depth 0.6 (Ø6.08 on the wall), centre X 25.6, Z 7.95 (S1 axis); full sphere, symmetric about the S1 axis (detail E-E)	depth +/-0.05
17	Knurl panel (bottom face)	X 40.5 to 81.5, 12.4 wide, corners R2.0: outline = V-groove frame (centre line), depth 0.35; inside diamond knurl: 90 deg V-grooves at +/-45 deg to X, pitch 1.6 (2.263 along X), depth 0.35, crossings at X 61.0 + k x 2.263, Y 0. Lid: same panel on its top face	depth +/-0.05 detail K
18	Tail window (bottom face)	5.2 x 14.0, depth 1.0, X 88.0 to 93.2; all corners R0.8, rear corners 1.8 x 45 deg; rim edge sharp in the STEP: break 0.2. Web to the bay corner 1.118 (window front X 88.0, bay end X 87.0)	depth +/-0.05
19	Lanyard slot, THRU	stadium 4.4 x 7.6 (R2.2 ends), centre X 90.2, Y 0, through tray and lid; walls: front 1.0, end 2.0; chamfer C0.35 on the lid top only (AUR-H-02), tray side edges: break 0.2	+/-0.1
20	Tail facets 2x	2.0 x 45 deg on both tail corners, full height; tail face outline bevel C0.8, front face outline chamfer C0.2	ISO 2768-mK
21	Marking field (bottom)	X 8.0 to 37.5, Y +/- 7.5; laser engraving through the anodise, artwork Aurora_marking_class2.dxf or _class3R.dxf (1:1), see note 6	position +/-0.2
22	Finish (all faces)	glass-bead blast, then black anodise type II, 8-12 um (note 5)	

NOTES
1 Material: aluminium 6061-T6. Geometry per Aurora_tray.step; sheet 1: interior, threads and fits.
2 General tolerances ISO 2768-mK; groove and knurl depths +/-0.05.
3 All V-grooves: 90 deg V-cutter or engraving tool, pyramid run-outs at the ends as the cutter leaves them, sharp apex. Cut the knurl and the grooves before blasting.
4 Do not cut deeper than shown: the walls left by these cuts are thin (note 7).
5 Finish: fine glass-bead blast, then anodise type II matte black (8-12 um). Knurl tips must stay sharp (rounding < 0.05).
6 Marking: laser-engrave through the anodise (bright aluminium). Artwork origin = middle of the front edge of the bottom face, X to the right, seen from below. Class 2 module (<= 1 mW): Aurora_marking_class2.dxf; Class 3R module (<= 5 mW): Aurora_marking_class3R.dxf.
7 Walls left by the exterior cuts (mm, from the STEP): 1.15 knurl and floor under the knurl; 1.2 side grooves to the cavity; 0.9 thumb dish to the cavity; 0.825 V-ring to window bore; 1.118 tail window to bay corner.
8 Details: V and E-E, F-F as marked; K is a schematic profile from the parameters. M2 hole (5) and everything inside: sheet 1.

MAIN BODY (TRAY) - EXTERIOR

Part: AUR-H-01 Aurora_tray.step	SCALE 2:1 (details as marked) SHEET 2/2
Aurora++ keychain flashlight v0.5 2026-10-14	Units mm | ISO 2768-mK